Towed-Sensor Surveillance UAV

AMERICAN UNIVERSITY OF THE MIDDLE EAST

1.0 EXECUTIVE SUMMARY

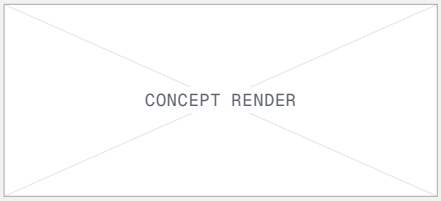
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DESIGN AT A GLANCE



WINGSPAN	M2 GROSS MASS
[span] m	[TOGW M2] kg
SENSOR MASS	M2 PAYLOAD
[m sensor] kg	[units] units
M2 FIVE-LAP TIME	M3 LAPS
[5-lap time] s	[laps] laps
GM DROP HEIGHT	PRED. MISSION SCORE
[h] m	[total] / 7.50

2.0 MANAGEMENT SUMMARY

2.1 Team Organization

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2.2 Schedule and Milestones

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TABLE 01 Sub-team responsibilities and the skill sets each requires.

SUB-TEAM	RESPONSIBILITY	REQUIRED SKILLS
Aero & Performance	Sizing, lap and energy model, stability and control	[skills]
Structures & Manufacturing	Airframe design, layup and build, load tests	[skills]
Propulsion & Avionics	Motor, propeller, battery, ESC, radio and fail-safe	[skills]
Sensor Systems	Sensor, D&R reel, shipping container, rigging	[skills]
Flight Operations	Pilot, observer, ground crew, checklists	[skills]
Business & Logistics	Budget, sponsorship, travel, documentation	[skills]

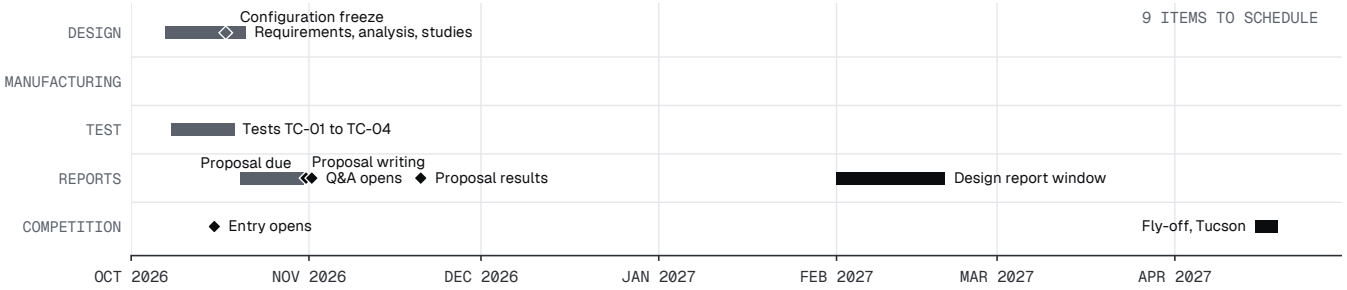


FIG. 01 [takeaway]. Milestones: proposal 31 Oct; Q&A rounds from 1 Nov; PDR; CDR; prototype first flight; sensor bench test; design report 19 Feb; competition aircraft first flight; Pre-Tech/First-Flight certification; fly-off 15–18 Apr.

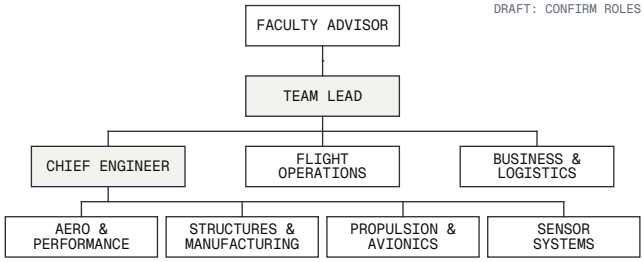


FIG. 02 [takeaway]: one owner for every scored subsystem.

TABLE 02 Program budget, all costs including travel (USD).

CATEGORY	COST	SHARE
Airframe, prototypes, test rigs	$1,800	[%]
Propulsion, avionics, radio	$1,914	[%]
Batteries, charging sacks, fuses	$1,029	[%]
Sensor electronics and lights	$547	[%]
Containers, simulators, drop rig	$245	[%]
Spares, props, adhesives, consumables	$357	[%]
Crating materials	$150	[%]
Travel and aircraft freight	[$]	[%]
Fly-off fee, AIAA/AMA memberships	[$]	[%]
Contingency ([%])	[$]	[%]
Total	[$]	**100%**

2.3 Budget

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TABLE 04 Mission requirements decomposed into subsystem requirements. Requirement IDs (SR) are cited in later trades and design decisions; † team requirement beyond the rules.

MISSION	SCORE	KEY CONSTRAINTS	DERIVED SUBSYSTEM REQUIREMENTS
M1 Staging	$M_1 = 1.0$	3 laps in 5 min; no payload; may be waived and credited by a successful M2	SR-01 static margin `[SM band]` % MAC in all load cases†; SR-02 battery, payload and control checks within the 5-min staging
M2 Delivery	$M_2 = 1 + \frac{W/t}{(W/t)_{\max}}$	Sensor in container plus simulators (±3.175 mm, ±28.3 g); TOP face up; 5 laps timed from first throttle-up	SR-03 bay for the container and N simulators; SR-04 positive mechanical restraint; SR-05 CG within marked limits in all load cases
M3 Sensor	$M_3 = 2 + \frac{NW_s}{(NW_s)_{\max}}$	Deploy before the first downwind turn; lights OFF, SOLID, FLASHING, OFF; stow before landing; sensor ≥ 152.4 mm (6 in), internal ballast	SR-06 internal, permanent powered reel, tow ≥ 1.5b, stowed sensor locked; SR-07 three lights and modes commanded over the tow line, visible in sun; SR-08 stable in tow; SR-09 own COTS battery (§3.2.3); SR-10 line clear of the propeller†
Ground	$GM = 0.5 + \frac{W_s h}{(W_s h)_{\max}}$	6 drops onto concrete, $h \le 1.524$ m (60 in), whole inches; ring to container bottom < 381 mm (15 in); 5-min rigging clock; zero damage	SR-11 container absorbs $W_s h$, no sensor damage; SR-12 tool-less O/E rigging, flush fittings; SR-13 test box drives all three modes via the tow connector
System	–	Span ≤ 1.829 m (6 ft); TOGW < 24.95 kg (55 lb); ≤ 100 Wh; no autopilot, GPS or camera	SR-14 pilot-in-the-loop only; SR-15 positive latches, no magnets; SR-16 energy for M2 at full payload; SR-17 §4.2.5 fail-safe on all outputs (gear down†), low-power range check; SR-18 structure for the race turn (n = `[n turn]`) and tip lift; SR-19 ESC and battery cooling†

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3.2 Scoring Model and Sensitivity Study

The total mission score is the sum of four mission scores, each a fixed completion credit plus a term normalized by the best team in the field:[1]

$$S_{\mathrm{M}} = M_1 + \left[1 + \frac{(W/t)}{(W/t)_{\max}}\right] + \left[2 + \frac{NW_s}{(NW_s)_{\max}}\right] + \left[0.5 + \frac{W_s h}{(W_s h)_{\max}}\right] \tag{1}$$

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FIG. 03 `[takeaway]`: which parameter moves total mission score most.

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[1] AIAA, *2026–27 Design, Build, Fly Rules*, Draft, Sep. 2026, Sec. 3.3.



FIG. 04 [takeaway]: the chosen design point on the coupled M2/M3/GM trade.

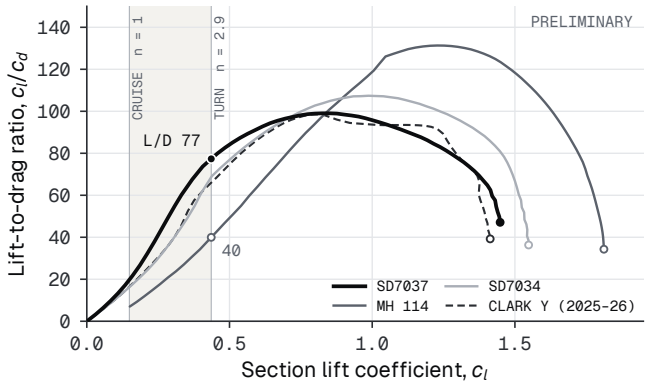


FIG. 05 [confirm W/S and V in airfoil.toml] Selected for the race envelope, not peak efficiency: in the turn, SD7037 gives 1.9× the c_l/c_d of MH 114 (77 against 40), the highest-peak section of the 13 tested. XFoil 6.99, $Re = 5.0 \times 10^5$, $N_{crit} = 9$; it over-predicts $(L/D)_{max}$ by 9–22 % against UIUC tunnel data. Curves end at $c_{l,max}$. Shaded: $c_l = 2n(W/S)/(\rho V^2)$ from cruise to the $n = 2.9$ turn at $\rho = 1.064$ kg/m³, the 1-in-10 hot afternoon at TIMPA.

3.3 Configuration Trade Studies

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3.4 Preliminary Design and Sizing

3.4.1 Sizing method

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TABLE 05 Configuration decisions and the requirement that drove each.

DECISION	OPTIONS CONSIDERED	SELECTED	DRIVER
Wing planform	High-AR, delta	[sel.]	[SR]
Airfoil	13 sections	[sel.]	[SR]
Wing position	Low, mid, high	[sel.]	[SR]
Tail	Conventional, V, H	[sel.]	[SR]
Landing gear	Tricycle, taildragger	[sel.]	[SR]
Retraction	Fixed, retractable	[sel.]	[SR]
Propulsion	Tractor, pusher, twin	[sel.]	[SR]
Sensor bay	Door, open slot	[sel.]	SR-06

TABLE 06 Deployment-and-recovery architecture trade (scores 1–5; selected option inverted).

CRITERION	WT.	OPT. A	OPT. B	OPT. C
Recovery reliability	[w]	–	–	–
Signal path integrity	[w]	–	–	–
Deploy/stow time	[w]	–	–	–
Weight and volume	[w]	–	–	–
Manufacturability	[w]	–	–	–
Weighted total	**1.00**	**–**	**–**	**–**

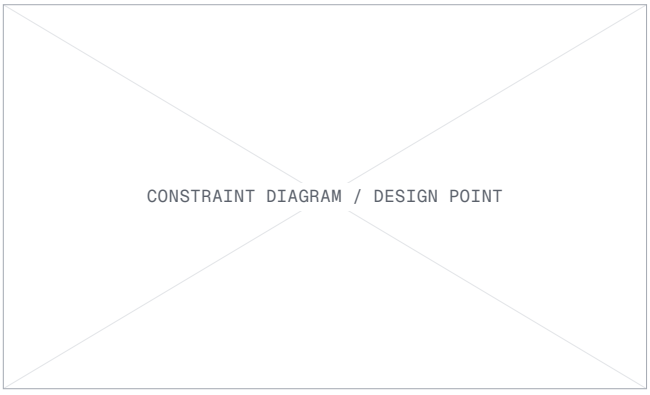


FIG. 06 [takeaway]: which requirement sets T/W and W/S.

3.4.2 Aircraft concept

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[2] 10th percentile at the daily high of 2,878 hourly KTUS observations, 15–18 April 1996–2025, reduced to TIMPA's 667 m (USGS 3DEP). Iowa Environmental Mesonet, "ASOS Data Download," Iowa State Univ., https://mesonet.agron.iastate.edu [retrieved 11 Oct. 2026].

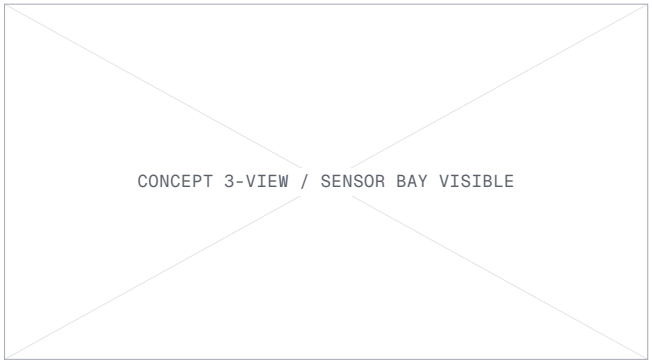


FIG. 07 [takeaway]: the configuration and where the sensor exits.

TABLE 07 Preliminary design parameters.

PARAMETER	VALUE	UNIT
Wingspan	[span]	m
Wing area	[S]	m²
Aspect ratio	[AR]	–
Empty mass	[m empty]	kg
Gross mass, M2 / M3	[TOGW M2] / [TOGW M3]	kg
Battery energy	[Wh]	Wh
Cruise speed	[V cruise]	m/s
Air density, hot day	1.064	kg/m³
Sensor mass (max)	[m sensor]	kg
Tow-line length	[tow length]	m

TABLE 08 Mission targets and predicted scores.

MISSION	TARGET	SCORE
M1 Staging	[waive or fly]	[M1]
M2 Delivery	[units] units, [m payload] kg, [5-lap time] s	[M2]
M3 Sensor	[laps] laps at [m sensor M3] kg	[M3]
GM Ground	[h] m at [m sensor] kg	[GM]
Total mission score		[total]

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3.4.3 Sensor and D&R system

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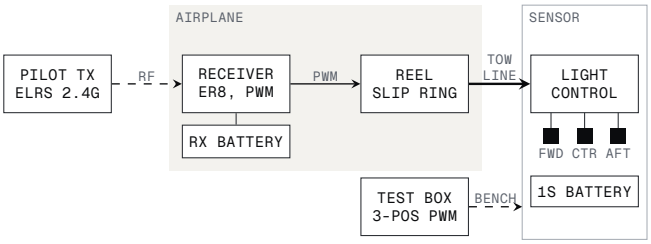


FIG. 08 [takeaway]: the command path to the sensor lights.

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3.4.4 Shipping container and rigging

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3.5 Mission Design Approach and Scoring Targets

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4.0 MANUFACTURING PLAN

4.1 Preliminary Manufacturing Flow

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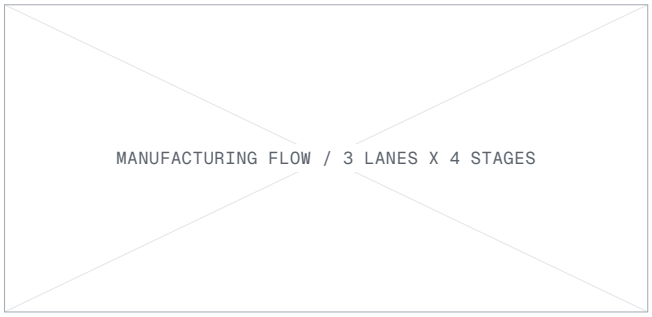


FIG. 09 [takeaway]: parallel airframe, sensor and container lanes converge at integration.

TABLE 09 Critical processes, why they matter, and risk mitigation.

PROCESS	WHY CRITICAL	MITIGATION
Reel slip ring	Signal and tow on one line	Bench cycling on a spare receiver
Sensor body	Stability, drop survival	Blower-rig screen (TC-01)
Container liner	Absorbs $W_s h$	Drop series to 1.524 m (TC-02)
Wing spar	Tip-load test at MGTOW	Prototype tip-lift proof test

4.2 Critical Processes and Technologies

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5.0 TEST PLANNING

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TABLE 10 Component, ground and flight test plan.

TEST	PASS CRITERION	DATE
Ground and component		
Thrust stand	Amps < fuse rating	[date]
Wing-tip lift at MGTOW	No damage (§4.2.4)	[date]
Light visibility in sun	≥ 90 % at 200 m	[date]
Tow-line signal	3 of 3 modes	[date]
D&R cycle	≥ 1.5b out, stows	[date]
Drop series to 1.524 m	No damage, 3/3 lit	[date]
GM drill, 5 min	Worst run ≤ 4:00	[date]
Staging drill, 5 min	Worst run ≤ 4:00	[date]
Range, fail-safe, CG	§4.2.5 checks pass	[date]
Flight		
Handling and trim	Level-flight video	[date]
Sensor tow stability	No rotation	[date]
Stepped M2 payload	5 laps < 5 min	[date]
Full M3 sequence	§3.3.3.3 sequence	[date]
Energy per mission	≥ 20 % reserve	[date]